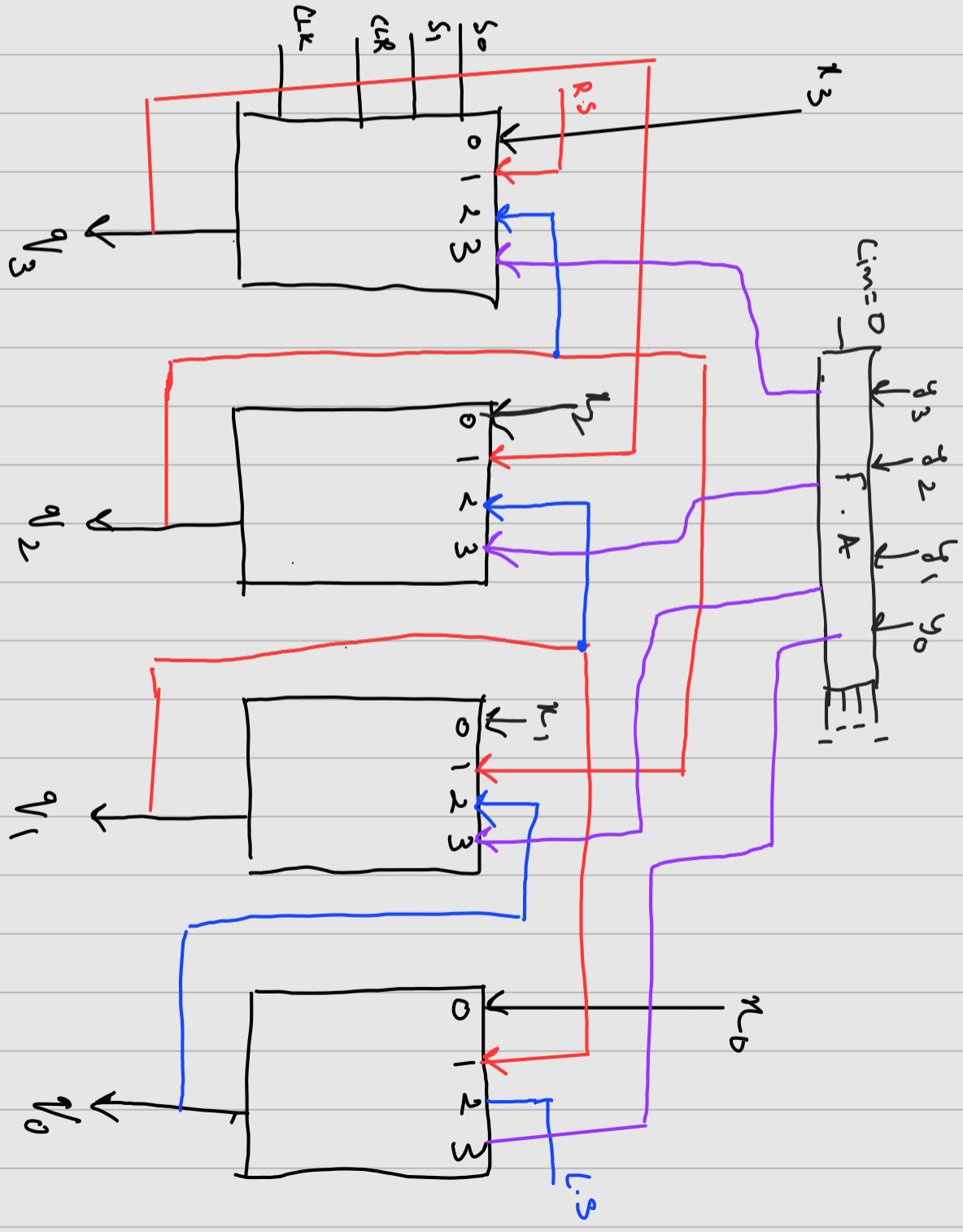
ACA Assignment -1

1000

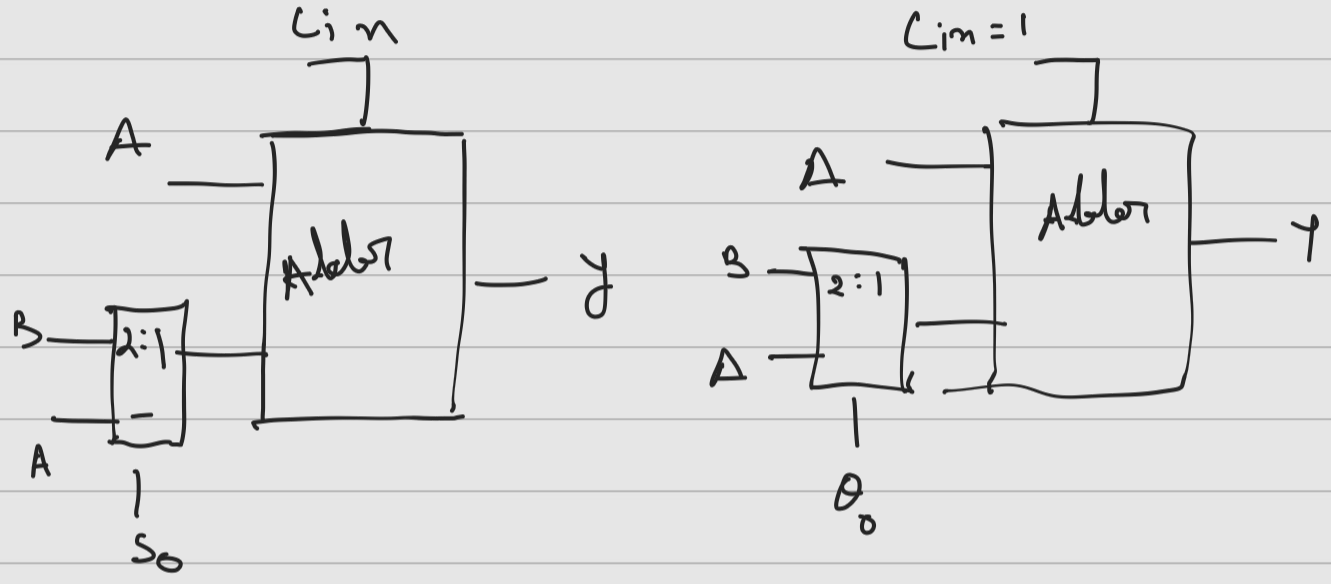
1] Design a general purpose register

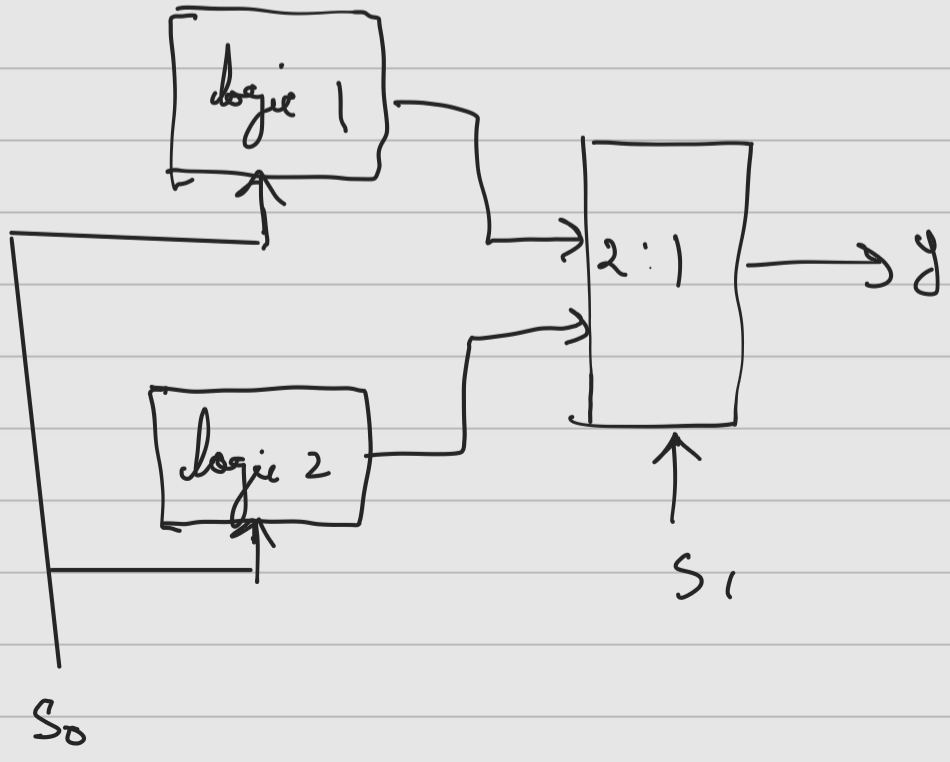
S_1	S_0	Operation
0	0	Enabled Data
0	1	Right shift
1	0	Left shift
1	1	Decrement



2]

S_1	S_0	F
0	0	A plus B
0	1	shift left (A) = A + A
1	0	A plus B' plus 1
1	1	shift left (A) + 1

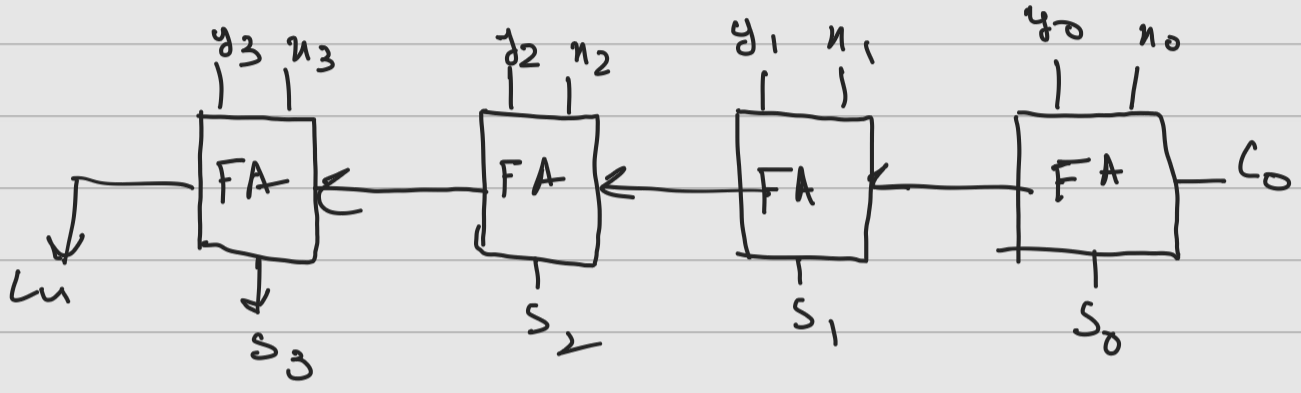




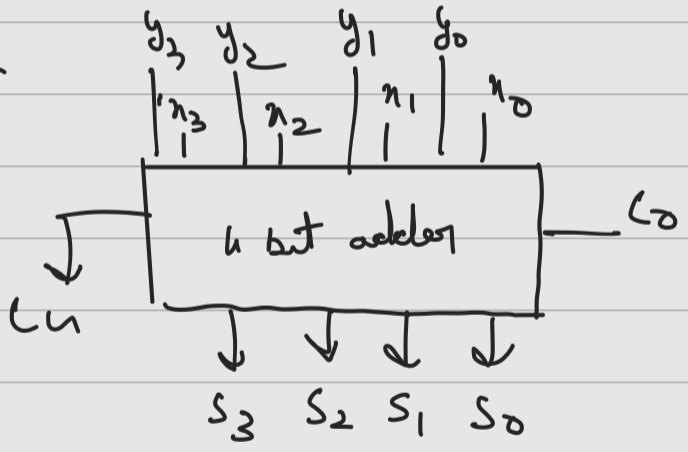
2) 16 bit adder using 4 bit adders.

-> Using Ripple Carry Adder

One 4 bit adder :-



Single Block

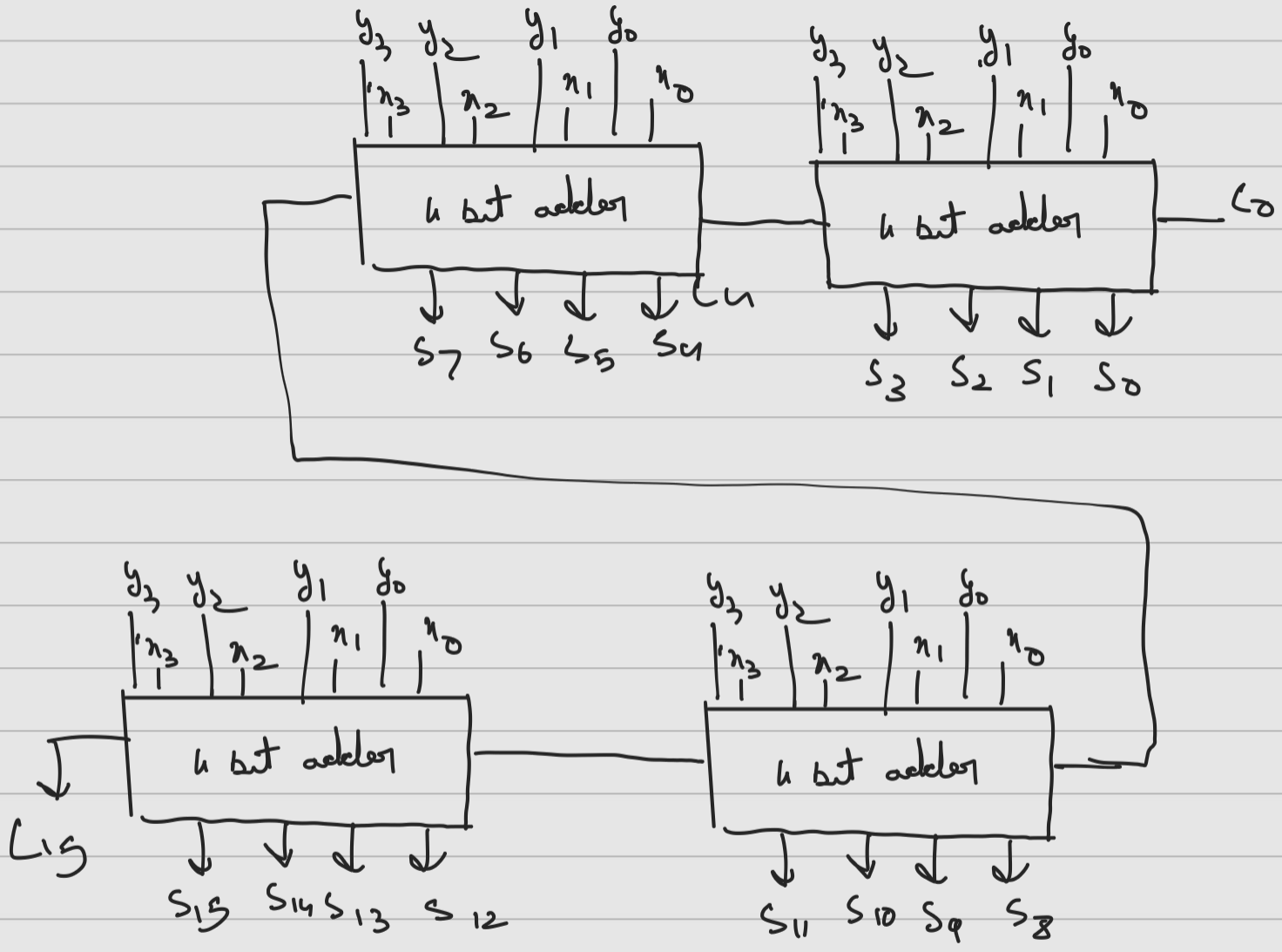


-> In worst case carry propagates through 15 adders

Delay $I_0 \rightarrow C_{15} = 15 \times 2\Delta$

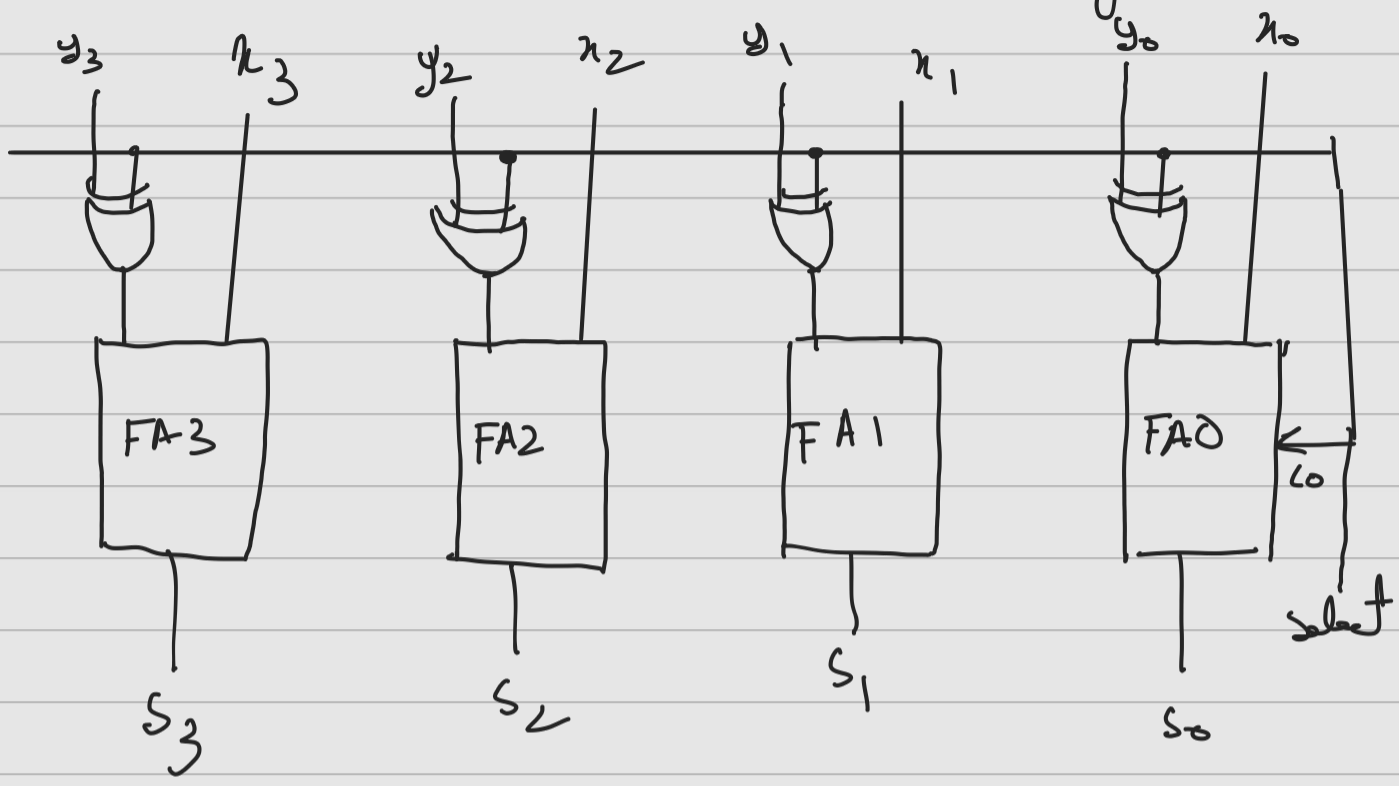
S_{15} from $C_{15} = 3\Delta$

Total Delay = $33\ \Delta$



16 bit Carry propagate adder

3) 4 bit 2's complement addition subtraction logic ckt



If select = 0, $C_0 = 0$ as $0 \oplus y_i = y_i$

So $x_i + y_i = S_i$

if select = 1, $C_0 = 1$ as $1 \oplus y_i = y_i'$

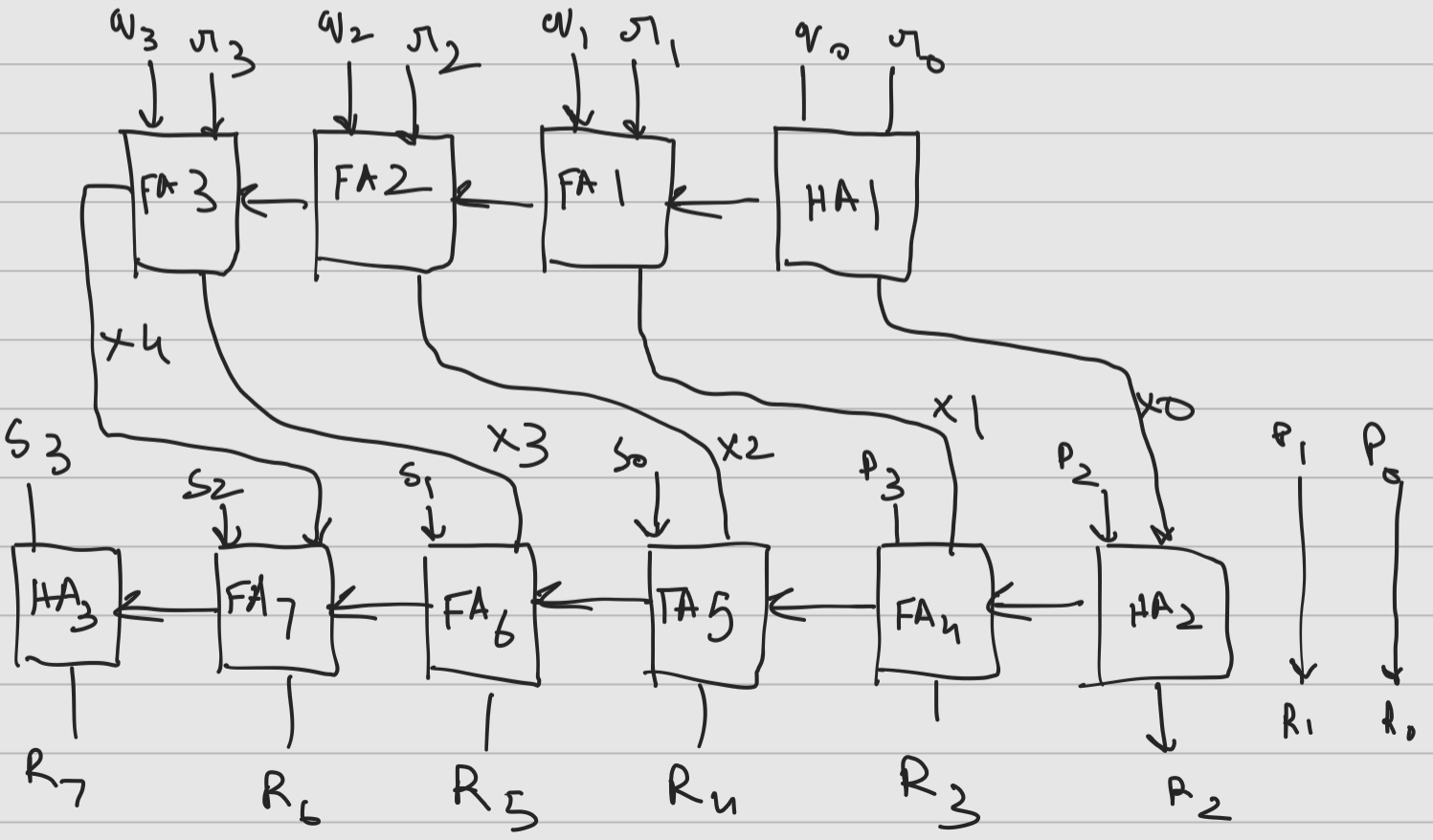
$S = x + y' + 1$

$= X + 2$'s complement of y

$\triangleq x - y$

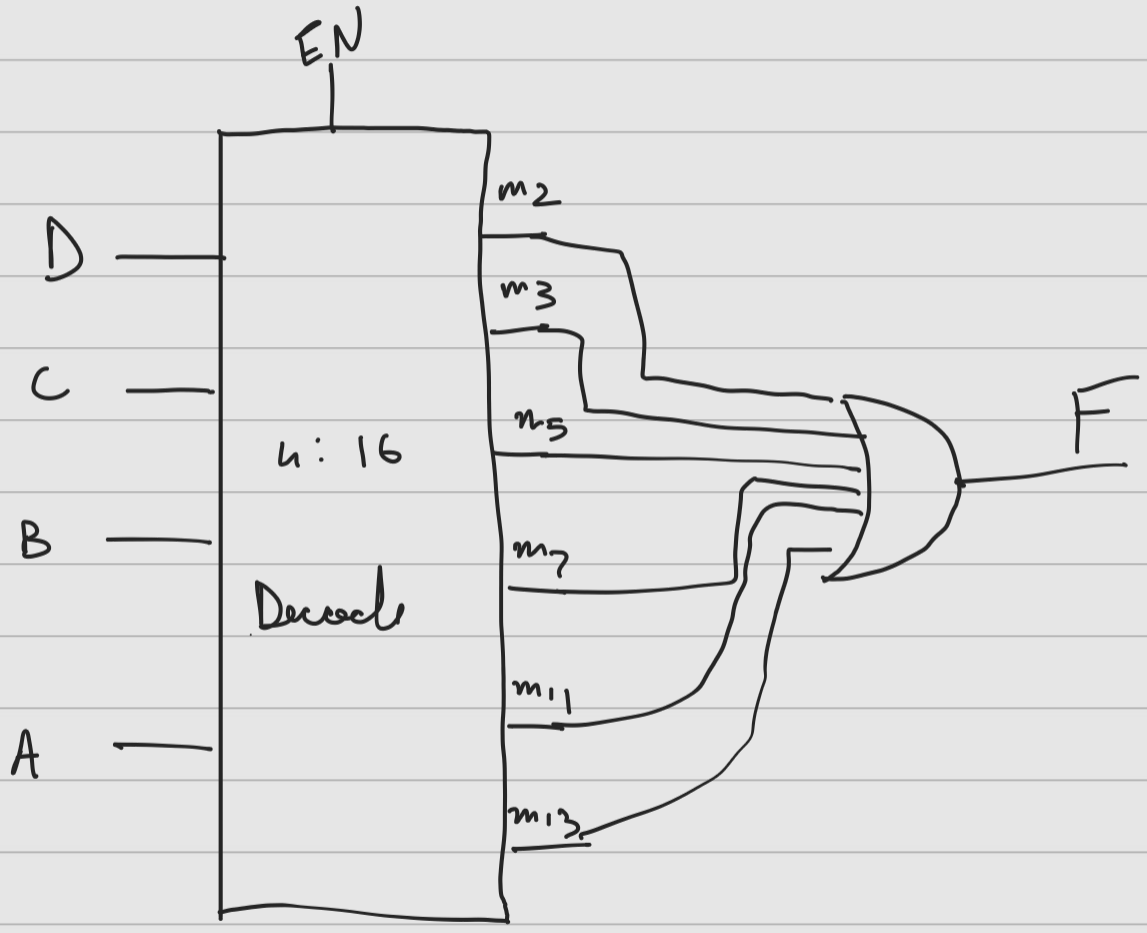
4) Use 4 2×2 ROMS each is 16×4

ROM 1	ROM 2	ROM 3	ROM 4
A[1:0] × B[1:0]	A[1:0] × B[3:2]	A[3:2] × B[1:0]	A[3:2] × B[3:2]
out $p_3\ p_2\ p_1\ p_0$	out $q_3\ q_2\ q_1\ q_0$	r_3, r_2, r_1, r_0	S_3, S_2, S_1, S_0



5) Input A B C D , Output F Boro or Not

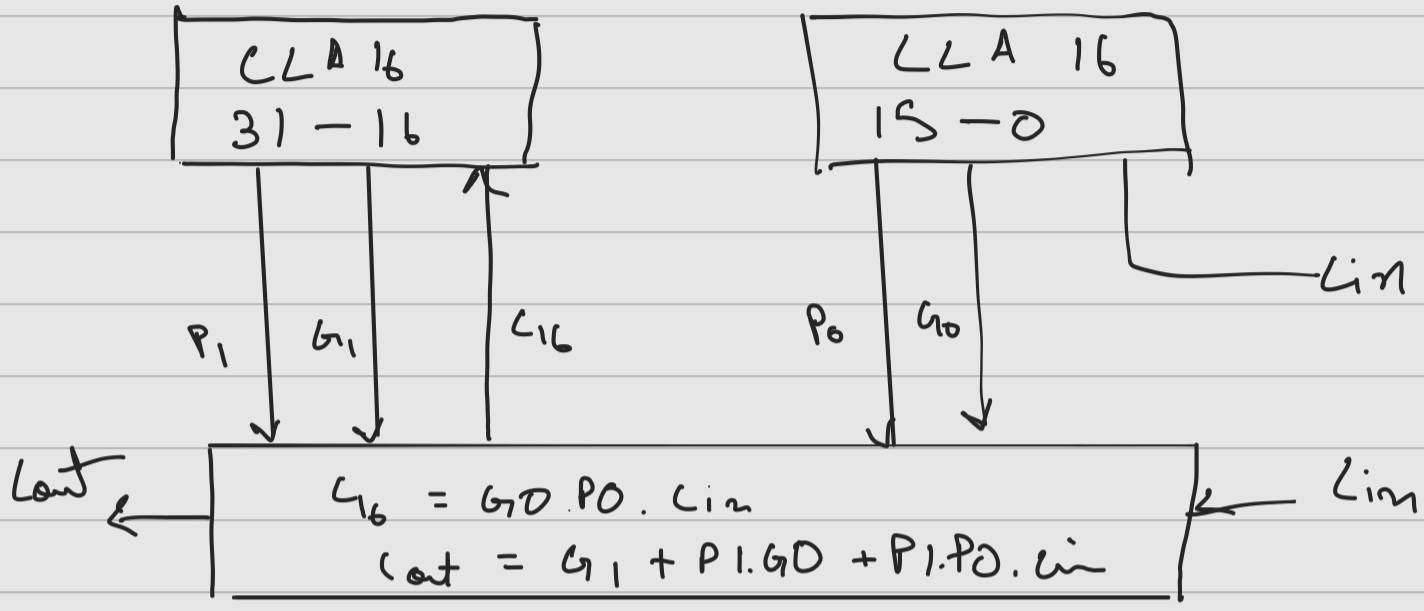
4 bit prime detector



if F=1 prime number else if F=0 not prime

6) 32 bit CLA, 64 bit CLA

32 bit CLA:



64 bit CLA

